

KANDALAKSHA, Russian Federation

WMO: 222170

Lat: 67.1692N

Lon: 32.3542E

Elev: 25

StdP: 101.02

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.9	-26.3	-32.3	0.2	-29.8	-28.7	0.3	-26.3	7.7	-6.6	6.7	-7.9	1.6	0	0.684

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.8	23.5	16.6	21.6	15.7	19.9	14.9	17.9	21.9	16.7	20.2	15.6	18.8	2.8	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.2	11.6	19.4	15.1	10.8	18.4	14.0	10.0	17.4	50.5	21.7	47.0	20.3	43.8	19.0	22.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.3	5.4	4.9	-32.3	27.4	3.7	2.2	-35.0	29.0	-37.2	30.3	-39.3	31.5	-42.0	33.1		
			-32.4	19.3	3.7	1.6	-35.1	20.4	-37.3	21.4	-39.3	22.2	-42.0	23.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	0.8	-11.9	-11.5	-6.5	-0.6	5.1	11.2	14.8	12.5	7.5	1.6	-4.8	-8.6
	DBStd	10.41	7.46	7.40	5.40	4.17	3.80	3.64	3.29	3.28	3.31	4.04	5.53	6.78
	HDD10.0	3684	679	602	512	319	158	28	3	11	88	262	445	577
	HDD18.3	6412	938	836	770	569	409	217	117	183	324	519	695	835
	CDD10.0	326	0	0	0	0	8	63	152	89	14	0	0	0
	CDD18.3	12	0	0	0	0	0	2	7	3	0	0	0	0
	CDH23.3	71	0	0	0	0	1	13	36	21	0	0	0	0
	CDH26.7	8	0	0	0	0	0	3	3	3	0	0	0	0
Wind	WSAvg	2.3	2.2	2.3	2.4	2.4	2.6	2.6	2.4	2.2	2.2	2.2	2.3	2.3
Precipitation	PrecAvg	509	35	26	29	24	37	52	66	59	55	50	44	33
	PrecMax	690	76	55	61	54	89	109	146	122	108	128	103	64
	PrecMin	362	6	8	4	1	6	9	0	19	10	7	7	8
	PrecStd	78	17	12	15	13	21	25	32	21	27	26	19	14
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.0	3.7	6.6	11.7	20.4	24.9	26.8	26.2	18.3	11.3	6.8	3.6
		MCWB	0.9	2.2	2.9	6.3	12.3	16.3	18.2	18.6	13.7	9.5	5.8	1.9
	2%	DB	0.7	1.3	3.7	8.5	16.4	21.4	24.0	22.4	15.9	9.8	4.7	2.2
		MCWB	0.1	0.3	1.2	4.4	9.9	14.5	16.8	16.3	12.2	8.3	3.9	1.3
	5%	DB	-0.6	0.1	2.0	6.6	13.9	19.2	22.3	20.0	14.3	8.3	2.9	1.0
		MCWB	-1.2	-0.7	0.3	3.0	8.6	13.1	16.3	15.6	11.5	7.2	2.3	0.4
	10%	DB	-2.0	-1.4	0.8	5.0	11.7	17.4	20.8	18.0	12.9	7.1	1.7	-0.1
		MCWB	-2.4	-2.1	-0.6	2.1	7.6	12.1	15.5	14.6	11.0	6.1	1.2	-0.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.2	2.3	3.1	6.8	12.6	17.7	20.1	19.7	14.3	9.9	5.8	2.5
		MCDB	1.6	3.6	5.9	10.8	18.2	23.0	23.8	24.8	17.0	10.8	6.5	3.3
	2%	WB	0.2	0.6	1.7	4.7	10.6	15.6	18.4	17.4	13.0	8.4	4.0	1.4
		MCDB	0.6	1.2	3.2	7.8	15.2	19.9	22.2	20.9	15.0	9.4	4.6	2.0
	5%	WB	-1.2	-0.6	0.6	3.5	9.1	14.0	17.2	16.0	12.0	7.3	2.4	0.5
		MCDB	-0.7	0.0	1.7	5.9	12.8	17.9	20.9	19.1	13.9	8.2	2.7	1.0
	10%	WB	-2.4	-2.0	-0.4	2.5	7.9	12.9	16.2	14.9	11.1	6.2	1.2	-0.6
		MCDB	-2.0	-1.4	0.8	4.5	11.4	16.5	19.5	17.6	12.5	7.0	1.6	-0.1
Mean Daily Temperature Range	5% DB	MDBR	7.2	7.8	8.7	8.1	7.4	7.8	7.8	7.6	6.8	4.8	5.0	6.3
		MCDBR	5.8	6.0	8.1	9.9	11.4	11.1	11.0	10.5	8.9	5.4	4.1	4.8
		MCWBR	5.6	5.4	6.3	6.3	6.2	5.5	5.1	5.5	6.0	4.3	3.7	4.3
	5% WB	MCDBR	5.6	5.9	7.0	8.7	10.3	9.1	9.1	8.9	7.5	4.8	3.9	4.5
		MCWBR	5.4	5.5	5.7	5.8	6.0	5.1	4.9	5.2	5.4	4.0	3.7	4.3
Clear-Sky Solar Irradiance	taub	0.209	0.269	0.304	0.338	0.356	0.355	0.374	0.353	0.323	0.286	0.358	N/A	
	taud	1.923	2.063	2.117	2.199	2.372	2.445	2.428	2.460	2.517	2.300	2.649	N/A	
	Ebn at Noon	139	535	727	809	840	851	818	798	740	536	171	0	
	Edh at Noon	13	53	87	106	100	96	95	83	61	42	15	0	
All-Sky Solar Radiation	RadAvg	0.06	0.52	1.93	3.56	4.58	5.12	4.66	3.27	1.77	0.58	0.11	0.00	
	RadStd	0.00	0.08	0.19	0.27	0.37	0.45	0.57	0.30	0.17	0.09	0.01	0.00	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (14 neighbors)	+0.60	N/A	N/A	N/A	N/A	N/A	-133	-200	N/A	N/A
		+0.55	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page